SRI SAI HEMANTH BOLLEPALLI

Master's student in Artificial Intelligence at the University of Technology Sydney, with a strong focus on applied deep learning, medical image segmentation, MRI analysis, and computer vision. Experienced in developing research-focused AI projects using nnU-Net, PyTorch, and model evaluation methods, including ongoing work on multi-region brain metastasis segmentation from multi-modal MRI. Interested in PhD research and broader-impact AI applications in medical imaging, neuroimaging, and clinically reliable deep learning systems.

PROJECTS

Multi-Region Brain Metastasis and Resection Cavity Segmentation — Ongoing Research

Developing a deep learning framework for automated segmentation of brain metastases and post-surgical resection cavities from multi-modal 3D MRI using customized nnU-Net v2 architectures. The project focuses on WT, TC, ET, and resection cavity segmentation across T1, T1c, T2, and FLAIR MRI modalities.

Rare-Class-Aware Multimodal Skin Lesion Diagnosis using ConvNeXt–EfficientNet Fusion

Developed a multimodal PyTorch pipeline for ISIC/MILK10k skin lesion diagnosis using paired clinical images, dermoscopic images, and metadata, with rare-class oversampling, ConvNeXt–EfficientNet ensembling, and Dockerized inference.

TRUST-ADMET: Trustworthy AI Framework for ADMET Prediction

TRUST-ADMET is an open-source Python framework for trust-aware ADMET prediction in drug discovery. It combines classical machine learning, graph neural networks, and molecular transformer models with reliability diagnostics to support safer, more interpretable molecular property prediction.

Automated Surface Defect Inspection System

This project is an end-to-end, deep learning-based industrial inspection system designed to automate metallic surface defect detection using state-of-the-art computer vision models.

RESEARCH INTERESTS

Clinical Decision Support Systems
Semantic Segmentation
Robust and Generalizable AI for Healthcare
Multi-modal MRI Analysis

CONTACT

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Synapse | Sage Bionetworks

EDUCATION

University of Technology Sydney 2025 – 2027
Master of Artificial Intelligence

Jawaharlal Nehru Technological University Kakinada 2020 – 2024
Bachelor of Technology in Computer Science

SKILLS

Programming Languages
Python, SQL

Artificial Intelligence & Medical Imaging
Deep Learning, Medical Image Analysis, Computer Vision, Brain Tumour Segmentation, Multi-modal MRI Analysis

Frameworks & Libraries
PyTorch, MONAI, nnU-Net v2, SimpleITK, NiBabel

Research & MLOps
Experimental Design, Model Development & Evaluation, Literature Review, Scientific Writing, Reproducible Research, Weights & Biases (W&B), ClearML

Tools & Platforms
Linux, Docker, CUDA, High-Performance Computing (HPC), Google Cloud Platform (GCP), Amazon SageMaker

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900) 2023

Google Cloud Architect Specialization, Coursera 2022